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Krupa-

1.Explain the various classes of data used for data processing with examples.

The various classes of data used are –

- Structured data
- Semi-structured data
- Unstructured data

<u>Structured data-</u>

When the data has a predefine schema or structure we call it as structured data. Or in simple words, The type of data that has a well-defined structure and can be stored in a relational database in the form of rows and columns. It follows a fixed schema, making it easy to store, query, and process.

Example- Customer records, Employee records etc.

Advantages:

- **Easy to store** because it follows a predefined structure.
- **Easy to query and access** using database query languages such as SQL.
- **Easy to process** because the data is organized into rows, columns, fields, and records.

Disadvantages:

- **Fixed schema:** The predefined structure makes it difficult to accommodate changes in the type or format of data.
- **Less flexibility:** Adding new fields or changing the existing structure may require changes to the database schema.
- **Not suitable for all types of data:** Images, videos, audio, free-form text, etc., cannot be represented efficiently.

<u>Semi-structured data-</u>

It is also referred as self-describing structure. The type of data which does not conform to a data model but has some structure. It is also called as the Quazi data. Does not have any rigid structure.

Sources- XML, JSON.

Examples- Emails, XML, HTML.

Advantages:

- **Flexible structure:** It does not require a fixed tabular schema.
- **Self-describing:** Tags or metadata provide information about the data.
- **Can represent hierarchical data** effectively.

Disadvantages:

- It does not have a **fixed schema**.
- **Processing is more difficult** than structured data.
- Relationships between data elements may not be clearly defined.

Unstructured data-

It does not conform to a data model, cannot be readily classified nor has a proper structure and does not fit into a neat box. It is also called as unclassified data. Indexing is not required here.

Examples-PowerPoint presentations, images, videos, letters etc.

Advantages:

- **Highly flexible:** It can contain text, images, audio, video, documents, social-media data, etc.
- Can store **large varieties of information** without requiring a predefined schema.
- Useful for extracting valuable information through **data mining, NLP and text analytics**.

Disadvantages:

- **Difficult to process** using traditional database systems.
- Does not have a **predefined data model**.
- **Difficult to organize and query**.

Prapthi

2.Discuss 3 V’s of Big Data and Challenges of Big Data

3 V’s of Big Data

Big Data is data that is big in Volume, Velocity and Variety.

1. Volume

Volume refers to the huge amount of data generated and stored.

Big data is generated from many sources such as:

Data storage and databases,Archives,Sensor data,Machine log data,Public,web,Social media,Business applications,Audio, video and images,Documents such as CSV, Word, PDF, XLS and PPT

Thus, the large volume of data creates a need for systems that can store and process huge datasets.

2..Velocity

Velocity refers to the speed at which data is generated, collected and processed.This includes real-time or near-real-time data streams, such as stock trades, IoT sensor readings, or social media feeds.

Data processing has changed from traditional batch processing to faster processing methods:

Batch → Periodic → Near real-time → Real-time processing

Examples include sensor data, online transactions and social media data, where information may need to be processed immediately.

3.Variety

Variety refers to the different types and formats of data. Big Data is mainly classified into three categories:

a) Structured Data:

Data obtained from traditional transaction processing systems and RDBMS. It is organized in a fixed format.

b) Semi-structured Data:

Data that does not follow a strict tabular structure but contains some organization. Examples are HTML and XML.

c) Unstructured Data:

Data without a predefined structure. Examples include text documents, audio, video, emails, photos, PDFs and social media.

Challenges of Big Data

1. Exponential growth of data

Data is growing at an exponential rate. The major challenge is to determine whether all the data can be analyzed and how useful knowledge can be separated from noise.

2.Cloud computing and virtualization

Managing Big Data requires suitable infrastructure. Cloud computing helps in managing Big Data through cost efficiency, elasticity and easy upgrading or downgrading. It also creates a challenge in deciding whether Big Data solutions should be hosted inside or outside the enterprise.

3.Data retention

It is difficult to decide how long Big Data should be retained. Some data is useful for long-term decisions, while some data may become irrelevant or obsolete shortly after it is generated.

4.Lack of skilled professionals

There is a shortage of skilled professionals with a high level of proficiency in data science, which is essential for implementing Big Data solutions.

5.Data processing challenges

Big Data creates challenges related to capture, storage, preparation, search, analysis, transfer, security and visualization. Traditional database systems may not be able to handle data that is too large, too fast and highly dynamic.

6.Data visualization

Large amounts of data make it difficult to present information in an understandable form. Therefore, data visualization has become an important discipline for representing Big Data effectively

LIKHITH RAI T

Q3. "It is easy to work with structured data compared to unstructured data." Justify with an example.

Definition

Structured data is data that is organized in a fixed and predefined format, usually in rows and columns. Examples include student records, employee records, bank transactions, and product information. Unstructured data does not follow a fixed format. Examples include photographs, videos, emails, voice recordings, and documents. Therefore, structured data is generally easier to store, access, search, update, and process compared to unstructured data.

Example

Consider a college student database:

Student ID	Name	Course	Semester	Marks
S101	Rahul	MCA	2	82
S102	Priya	MCA	2	91

The information is arranged in fixed columns. If we want to find all MCA students, update Rahul's marks, or delete a student record, the database can perform these operations easily.

In comparison, unstructured data such as a student's photograph, a recorded lecture video, an email, or a voice recording contains useful information but does not have a fixed row-and-column format. Therefore, organizing and searching specific information from it can be more difficult.

Ease of Working with Structured Data

1. Insert, Update and Delete

Structured data can be easily inserted, updated, and deleted using database operations. For example, we can add a new student, change a student's marks, or delete a student's record. This is easy because the information is stored in predefined fields.

2. Security

Structured data can be protected by giving different users different permissions. For example, students can view their own details, while authorized staff can modify marks or other sensitive information.

3. Indexing

An index helps the database find information quickly. For example, if we want to find all students from the MCA department, an index on the Course field can help retrieve the records quickly.

4. Scalability

Structured data can be managed even when the amount of data becomes very large. For example, a college database can grow from hundreds of student records to thousands or more, and additional storage and processing capacity can be provided when required.

5. Transaction Processing – ACID

Structured databases, particularly relational databases, support ACID properties, which help transactions remain safe and reliable.

- **Atomicity:** A transaction happens completely or not at all.
- **Consistency:** Data remains correct after a transaction.
- **Isolation:** Transactions occurring at the same time do not interfere with each other.
- **Durability:** Once data is saved successfully, it remains saved even if the system crashes.

4..Explain unstructured data ?How to deal with it? Explain techniques used to find unstructured data?

Unstructured data is data that does not follow any predefined data model, fixed format, or organized structure. Unlike structured data stored in tables with rows and columns, unstructured data is difficult to store, process and analyze using traditional database techniques. The structure of unstructured data is often unpredictable, although some useful information or structure may be implied.

How to deal with it?

Today unstructured data constitutes approximately 80% of data that is being generated in any enterprise.

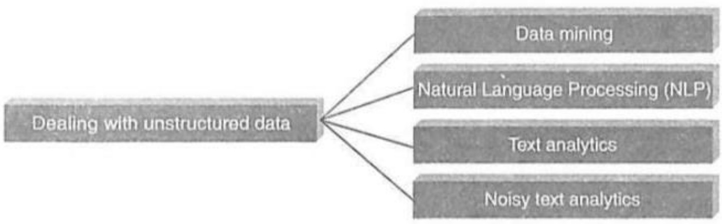

Figure 1.11 Dealing with unstructured data.

The following techniques are used to find patterns in or interpret unstructured data:

1.**Data mining**: It deals with large data sets and uses methods from artificial intelligence, machine learning, statistics, and database systems to find consistent patterns and relationships between variables. It is the analysis step of the “knowledge discovery in databases” process.

- Association Rule Mining: Also called market basket analysis or affinity analysis. It determines “What goes with what?” Example: buying bread may lead to buying eggs or cheese.
- Regression Analysis: Predicts the relationship between two variables. The variable to be predicted is the dependent variable, while the variables used for prediction are independent variables.
- Collaborative Filtering: Predicts a user’s preferences based on the preferences of a group.

2. **Text Analytics or Text Mining:**Text is largely unstructured, amorphous, and difficult to deal with algorithmically. Text mining is the process of extracting high-quality and meaningful information by finding patterns and trends from text. It includes text categorization, text clustering, sentiment analysis, and concept/entity extraction.

3**. Natural Language Processing (NLP):**It is related to human-computer interaction. It enables computers to understand human or natural language input.

4**. Noisy Text Analytics:**It is the process of extracting structured or semi-structured information from unstructured data such as chats, blogs, wikis, emails, message boards, and text messages. Noisy data may contain spelling mistakes, abbreviations, acronyms, non-standard words, missing punctuation, missing letter case, and filler words such as “uh” and “um”.

5. **Manual Tagging with Metadata:**It involves manually tagging unstructured data with adequate metadata to provide the required semantics for understanding the data.

6. **Part-of-Speech (POS) Tagging:**Also called grammatical tagging, it is the process of reading text and tagging each word according to its part of speech, such as noun, verb, adjective, etc.

7**. Unstructured Information Management Architecture (UIMA):**UIMA is an open-source platform from IBM used for real-time content analytics. It processes text and other unstructured data to find latent meaning and relevant relationships buried within the data.

-Pranamya

5. Difference between Traditional BI and Big Data.

Traditional Business Intelligence (BI)	Big Data
1. In a traditional BI environment, **all the enterprise's data is housed in a central server**. The database server typically **scales vertically**.	1.In a Big Data environment, **data resides in a distributed file system**. The distributed file system **scales horizontally (scaling in or out)**.
2. In traditional BI, data is generally analyzed in an **offline mode**.	2.In Big Data, data is analyzed in **both real-time as well as offline mode**.
3. Traditional BI deals with **structured data**.	3.Big Data deals with **variety of data: structured, semi-structured, and unstructured data**.
4.In traditional BI, **data is taken to the processing functions** — *move data to code*.	4.In Big Data, **processing functions are taken to the data** — *move code to data*.